Module 4

Swaps

September 17 & 22, 2025

Interest Rate Swaps (IRS)

Contract:

Exchange interest payment at pre-specified time intervals with pre-specified fixed rate and reference floating rate on a pre-specified notional amount.

- Common reference floating rate: SOFR/IBOR (e.g., EURIBOR)
- Swap legs: fixed leg, floating leg

Example:

Assume notional = N, floating reference rate = R_M, fixed rate = R_S ("swap rate"), interest exchange semi-annually at $T_1, T_2, \ldots, T_n$. At time T_i:

- fixed rate receiver: receives $NR_S(T_i - T_{i-1})$, pays $NR_M^{i-1,i}(T_i - T_{i-1})$
- fixed rate payer: pays $NR_S(T_i - T_{i-1})$, receives $NR_M^{i-1,i}(T_i - T_{i-1})$

$R_M^{i-1,i}$ = realized market rate for $[T_{i-1}, T_i]$

- R_M = IBOR, Term SOFR: rate is known at the beginning of the period T_{i-1}
- R_M = SOFR: rate is not known until the end of the period T_i

Overnight Indexed Swap (OIS): floating rate is computed from realized overnight rate

Rational for IRS:

- Matching asset/business income and liability
- Hedging interest risk
- Speculations on rates

IRS has become so liquid it becomes fundamental blocks of yield curves building

Valuation:

Assuming receiving fixed rate R_S. PV of all future cash flows of an IRS

$IRS = N\sum_{i=1}^{n}(R_S - R_M^{i-1,i})(T_i - T_{i-1})DF_i$, DF_i = discount factor from time T_i

- Pricing IRS as a series FRA:

  $$IRS = N\sum_{i=1}^{n}(R_S - R_M^{i-1,i})(T_i - T_{i-1})DF_i$$

  Each item in the above summation is a FRA with forward (strike) rate = R_S

  But why bother with IRS (instead of just a series of FRA)?

  - administrative convenience
  - transaction cost

- Pricing IRS as exchange of cash flows of a fixed rate coupon bond and a floating rate coupon bond:

  $$IRS = N\sum_{i=1}^{n}R_S(T_i - T_{i-1})DF_i - N\sum_{i=1}^{n}R_M^{i-1,i}(T_i - T_{i-1})DF_i$$

  $$= N\sum_{i=1}^{n}R_S(T_i - T_{i-1})DF_i + NDF_n - N\sum_{i=1}^{n}R_M^{i-1,i}(T_i - T_{i-1})DF_i - NDF_n$$

  $$= B(R_S, T_n, N) - FRN(T_n, N)$$

  $B(R_S, T_n, N)$ = fixed rate coupon bond with cpn rate R_S, maturity T_n, notional N

  $FRN(T_n, N)$ = FRN with maturity T_n, notional N

Swap rate at IRS inception (equilibrium swap rate) t = 0: IRS = 0:

Let $R_F^{i-1,i}$ = forward rates for $[T_{i-1}, T_i]$ at t = 0,

$$R_S^e = \frac{\sum_{i=1}^{n}R_F^{i-1,i}(T_i - T_{i-1})DF_i}{\sum_{i=1}^{n}(T_i - T_{i-1})DF_i}, \quad \text{“on-market”}$$

Note: IRS = 0 but each individual FRA can either be positive or negative

As t > 0: off-market $R_S \neq R_S^e$

Variations in IRS Contracts:

- Changing notional
- Different payment frequencies for fixed and floating legs
- Floating leg with spread, with caps, floors
- Floating-floating (basis) swaps
- Cancellable swaps

Remarks on IRS

What is "risk free" interest rate?

- Need be default risk free
- Need be accessible to market participants
    - TSY: accessible only by US government
    - OIS, SOFR: secured short lending/borrowing rate → "risk free"
    - IBOR: proxy to inter-bank lending rate → "risky"

      (EURIBOR, BSBY, Ameribor, …)

Market practice of "risk free" discounting:

- Before 2008 financial crisis, LIBOR is widely used to build "risk free" discount curve
- Post crisis OIS and other "secured rates" gradually replaced LIBOR for valuing collateralized derivative instruments (but not everywhere)

  → Practice of "differential discounting": discount rates depend on the terms of collateral agreement (e.g., credit support annex (CSA))

Valuation of IRS (& FRA)

- Floating reference rate and discount rate may be different (different curves, different compounding conventions)
- Using floating reference rate as discount rate: forward rates may be bootstrapped from the discount curve as discussed in the above
- Using "secured" (e.g., OIS) discount rate pricing a swap with "unsecured" reference floating rate (e.g., IBOR): IBOR forwards may be extracted from the prices of traded swaps

Example: An IRS that exchanges IBOR with fixed rate = 3% semiannually has 15 month remaining. Its notional is $100M. The 6mo IBOR fixed 3 month ago is 3.5%. The IBOR forward for 3 to 9 month and 9 to 15 month are 3.75% and 4.25%. The 3, 9, 15mo OIS rates (continuous compounding) are 3%, 3.25% and 3.5% respectively. The value of the IRS to the fixed rate receiver is -$1.2123M.

N	100				
Month	IBOR	Fix	OIS (CC)	DF	PV
3	3.50%	3.00%	3.00%	0.9925281	-0.2481
9	3.75%	3.00%	3.25%	0.9759197	-0.3660
15	4.25%	3.00%	3.50%	0.9571932	-0.5982
				Sum PV	-1.2123

Currency Swaps

Contract:

Exchange principal and interest payments in one currency for principal and interest payments in another currency. The principal is changed back to the original currency at the maturity of the contract.

Interest payments may be fixed or floating on either or both legs: fixed-fixed, fixed-floating, floating-floating.

Example:

Notional = $N_{\$}$, $N_{€}$, $ fixed rate r_d , € fixed rate r_f , semi-annual interest exchange.

Principal exchange at T_0:

$N_{€} \leftrightarrow N_{\$}$ (usually $N_{€} = FX_0 N_{\$}$)

interest exchange semi-annually at $T_1, T_2, \ldots, T_n$:

$N_{\$} r_d (T_i - T_{i-1}) \leftrightarrow N_{€} r_f (T_i - T_{i-1})$

Principal exchange at T_n:

$N_{\$} \leftrightarrow N_{€}$

Rational for currency swap:

- Matching asset/business income and liability
- Comparative advantage

Valuation:

- Pricing a currency swap as a long/short bond portfolio
- Pricing a currency swap as a series of FX forwads

Example: Hull 7.9 (p168-170)

More on IR and FX swaps:

See the reference cited at the end of lecture notes for Module 2

Total Return Swaps (TRS)

Contract:

Exchange total return (i.e., price change + applicable cash flows) of an asset with fixed or floating interest payments at pre-specified time intervals. It may also be referred as *equity swap*.

Example:

Fixed notional \$N, reference asset S, reference floating rate = R_M, swap frequency = 3mo

Asset return receiver: receives total return on S, pays interest on 3mo R_M at time T_1, T_2, …., T_n . If S pays no dividend the cash flow at T_i is

$$\left(\frac{S_i}{S_{i-1}}-1\right)N - R_M^{i-1,i}\left(T_i - T_{i-1}\right)N$$

Rational for TRS:

- Exposure to asset return without owning the asset
- Transfer risk characteristics
- Tax considerations

Valuation:

- Borrow \$N at time T_0, T_1, T_2, …., T_{n-1} at the prevailing reference floating (risk free) rate R_M and invest in S. This replicates TRS:

  At T_{i-1} one buys N/S_{i-1} shares at price S_{i-1} . At T_i its value is NS_i/ S_{i-1}, return for the period is $NS_i/ S_{i-1} – N S_{i-1}/S_{i-1} = N(S_i/ S_{i-1} – 1)$. The interest to be paid for N borrowing is $R_M^{i-1,i}\left(T_i - T_{i-1}\right)N$, hence the net cash flow

  $$\left(\frac{S_i}{S_{i-1}} - 1\right)N - R_M^{i-1,i}\left(T_i - T_{i-1}\right)N$$

  At $t = T_0 = 0$, TRS = 0:

  → "fair" floating rate to be paid for asset return is the reference floating rate R_M (e.g., IBOR), assuming R_M can be accessed by both parties

  Arbitrage when the floating rate $\neq R_M$:

  Assume the interest leg of TRS is R_M + s (s = spread > 0): replicate TRS as above with interest cost = R_M , enter the TRS that pays asset return and receives R_M + s , net risk free cash flow = $sN(T_i – T_{i-1})$ per period.

  R_M seems "cheap" for exchanging on the asset return …. But really?

- Fixed Notional \$N, swap fixed rate interest income with asset return:

  TRS with fixed interest leg = TRS with floating interest leg + IRS

  → "fair" rate for the fixed interest leg = IRS rate

  → engineering a derivative product with two other derivatives

Variations in TRS Contracts:

- Changing notional: e.g., notional compound with asset return
- Exchange return on one asset with return on another
- Cross currency TRS
- Cancellable swaps (one-sided or two-sided)

An example of misuse of TRS:

- Collapse of Archegos Capital Management (March 2021)

  https://www.wsj.com/articles/what-is-archegos-and-how-did-it-rattle-the-stock-market-11617044982

  → it is also an example of mismanagement of counter-party risk

Credit Default Swaps (CDS)

Contract:

Exchange a stream of pre-specified cash flow for credit default protection.

- The reference assets
- The notional of the CDS
- The protection period
- The CDS spread (rate of protection) and payment frequencies
- The definition of credit events
- The payment at the credit event and the settlement method (cash/physical)

Example:

- $10m 5 year CDS on a 20 year corporate bond with 6.25% (625 basis points) CDS spread payable semi-annually. Credit events include i) bankruptcy filling by the issuer, ii) missed coupon/principle payments; iii) obligation acceleration; iv) repudiation/moratorium; ….. Cash settlement.

Variations in CDS Contracts:

- Single name CDS
- CDS on a basket of assets:
  - First to default (... n-th to default) protection
  - Protection on a specific portion ("tranche", e.g., equity, mezzanine, and senior tranches) of the basket
- CDS indices (e.g., CDX.NA.IG, CDX.NA.FIN, iTraxx series)

- CDS with non-standard payoffs

Rational for CDS:

- Hedging credit risk

  Example: long debt, long CDS

- Directional trading & speculations

  Examples:

  - "The Greatest Trade Ever", Gregory Zuckerman
  - "'London Whale' Rattles Debt Market" (2012): https://www.wsj.com/articles/SB10001424052702303299604577326031119412436, https://archive.nytimes.com/dealbook.nytimes.com/2012/06/28/jpmorgan-trading-loss-may-reach-9-billion/london

- CDS/CDS indices may be served as a leading indicator for equity markets

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Remarks:

Derivatives such as forwards, futures, vanilla IRS, CS, TRS considered so far can be *statically* replicated/hedged. Not so for options. For such type instruments one needs *dynamical* replication/hedging, which we will consider next.